

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)	(i)		Maximum [or fastest, highest] speed [or velocity] /constant speed [when forces are balanced] Accept: <u>speed</u> when forces are balanced	1			1		
		(ii)		Indicative content: Preparation 1. Set up the apparatus as shown, with the pointer (e.g.) 150 cm above the floor 2. Use a balance to determine the mass of a cake case [e.g. weigh 10 and divide by 10] Carrying out 3. Drop a single cake case from (e.g.) 20 cm above the pointer. 4. Using a stopwatch, record the time it takes to fall from the level of the pointer to the floor. 5. Repeat step 4 (another 4 times) and calculate the mean time. 6. Repeat steps 4 and 5 with a stack of e.g. 2, 3, 4 [and 5] cake cases. Analysis 7. Use the equation $\text{speed} = \text{distance}/\text{time}$ to calculate the speed 5 – 6 marks Expect a full description of the method with at least 5 points mentioned from all areas. <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i>	6			6		6